AI Based Body Measurement App
Mobile APP

Final Year Design Project Proposal

by

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Mobile APP (V 1.0)

Abstract of Proposal

This application will be an AI-based solution designed to make custom-fitted online retail shopping convenient. It will address the time-consuming and error-prone nature of traditional body measurement methods by allowing users to input measurements either manually or through **AI-based camera key point tracking**. The core objective is to develop a system that accurately extracts body measurements from static, user-uploaded images using advanced models like Move-Net and **ML Kit**. All measurement calculations and model execution will perform directly on the user's device using **edge computing** for faster processing and improved data privacy.

Key features will be a real-time camera feed, **Pose Estimation** using the Pose-Net model integrated via TF Lite, and Firebase Integration for user authentication, real-time database storage, and cloud storage. Based on the calculated body measurements, the app will provide a tailored experience, allowing users to choose through different designs, texture and prints, a totally customizable dress, and providing recommendations for dress styles and sizes. We aim to verify the app to achieve almost **90% correctness** in body measurement output and **100% correctness** in data privacy compliance through local processing and storage.

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1. Introduction

1.1. Background

The development of this real-time body measurement application addresses critical challenges in the domains of fitness, fashion, and healthcare. Traditional methods for obtaining body measurements are often time-consuming, prone to human error, and require specialized tools or professional assistance. The integration of accurate body measurements with personalized clothing recommendations has been limited, leading to inefficiencies and suboptimal user experiences in online shopping. The app aims to serve as a pre-data collection tool for more automated systems like online shopping, virtual clothes selection, and automated tailoring, while utilizing edge computing making device run model make calculation independent of cloud processing for security and speed.

1.2. Problem Statement

In the fashion and retail industry, obtaining precise body measurements is a challenging and time-consuming task for both consumers and retailers. Traditional manual measurement techniques are highly prone to human error and inconvenience. The rise of online shopping has worsened the issue, as customers cannot physically try on clothes, leading to frequent sizing problems and product returns. There is a growing unmet need for personalized clothing that precisely adapts to individual body shapes, and many consumers are frustrated by the inefficiencies in current measurement and fitting practices.

1.3. Stakeholders & Interests

Stakeholder Type	Interest in the Solution	Potential Market Size
Online Shoppers	Seeking a more accurate and convenient way to determine their correct size without needing to try garments on physically.	Individuals who frequently purchase clothing online.
E-commerce Platforms	Enhancing customer experience by integrating intelligent sizing tools, reducing return rates, and improving overall satisfaction.	Online retail businesses.
Fashion & Apparel Brands	Aiming to deliver personalized shopping experiences by offering better-fitting clothing, reducing sizing-related returns,	Brands aiming to cater more effectively to diverse body types.

	and improving brand loyalty.	
Tailors & Custom Clothing Services	Requiring detailed and accurate body measurements for bespoke garments, offering a streamlined and remote alternative to in-person fittings.	Tailoring professionals and custom clothing services.
Health & Fitness Enthusiasts	Wishing to monitor body changes for wellness or fitness purposes and assessing progress over time.	Users who track body measurements over time.
Developers & Technologists	Interested in exploring or integrating body measurement features into their own applications, given the app's modular design.	Tech-savvy users and developers.

1.4. Objectives

Following is a list of the main objectives of the App to be designed:

- **Accurate Measurement Extraction:**
  To develop a system capable of accurately extracting body measurements from static, user-uploaded images using advanced AI models, eliminating the need for manual input.
- **Edge Computing for Local Processing:**
  To utilize edge computing by executing the model directly on the user's device. This reduces dependency on internet connectivity, increases processing speed, and improves data privacy.
- **Enhanced User Experience:**
  To create a clean, intuitive interface where users can easily upload their images and receive measurements with minimal effort.
- **Integration with Retail Platforms:**
  To connect measurement outputs with online shopping platforms, enabling personalized size and style recommendations.
- **Privacy and Data Protection:**
  To implement robust security measures that ensure all user data—including images and measurements—is handled responsibly and stored in accordance with modern privacy standards.

1.5. Scope

- **Inclusions**
  AI-based body measurement from user-uploaded images, personalized dress design recommendations using machine learning models, Flutter-based cross-platform mobile interface, TensorFlow Lite (TFLite) / ML Kit integration for edge (offline) pose detection, Firebase backend

services for authentication and data storage, and adherence to standard data security and privacy protocols.

- **Exclusions/Limitations:**
  The system is currently not desktop supported. Measurement accuracy may be affected by poor lighting conditions or incorrect user posture. The model is designed for single-person measurement only and does not include payment gateway integration, delivery tracking, or e-commerce features.

2. Literature Review

2.2. Related Work

- **Body Shape Classification (2025)**
  Supervised ML accurately classifies body shapes and extracts measurements for apparel fitting.
- **ML Clothing Size Prediction (2025)**
  Models like SVM and PCA-SVM predict clothing sizes from 3D scans with ~90% accuracy.
- **Fashion Digitization Evaluation (2025)**
  3D scanners outperform smartphones for precision tailoring, though phones are improving.
- **3D Body Scanning via Smartphone (2024)**
  New 3D avatar reconstruction methods offer accurate body shape and composition data using only a phone camera.
- **Smartphone Body Measurement (2021)**
  Smartphones can capture body size measurements with reasonable accuracy, making digital anthropometry affordable.
- **FITME ML Model (2019)**
  Machine learning helps estimate body measurements without manual tools.

Market Solutions
Tools like Zozosuit, 3DLook, and Bodygram automate measurements but still struggle with perfect accuracy and e-commerce integration. Their disadvantages are that they rely on manual measurement techniques, which are prone to human error, time consuming, and struggle with seamlessly integrating accurate measurements into personalized clothing recommendations in the e-commerce space.

- **Amazon BodyFit AI** – Size prediction using phone images + purchase data.
- **Walmart Zeekit** – Virtual fitting rooms using ML.
- **Google BodyShape Project** – Phone-based 3D body reconstruction.
- **Sizer Technologies** – Smartphone-based measurement for retail.
- **Bold Metrics** – AI-based clothing size recommendations.

2.2. Gap Analysis:

- **Accuracy and Convenience:** The app aims to provide **highly accurate measurements** (92% accuracy validated) through automated AI (MoveNet/ML Kit) and a convenient user interface, overcoming the manual techniques that are prone to human error and inconvenience.

- **Edge Computing/Offline Capability:** The system directly addresses the reliance on cloud processing by implementing **edge computing** for all model execution and measurement calculations locally on the device. This enhances speed, privacy, and allows the core measurement function to operate without continuous internet connectivity.
- **Seamless E-commerce Integration:** The app closes the gap between accurate measurement and informed fashion choice by enabling users to select and visualize dress designs based on their measurements, providing a practical solution to sizing issues and product returns in online shopping.
- **Data Security:** The reliance on local processing and storage ensures the app maintains **100% correctness** in following data privacy policies, a critical gap for sensitive body data.

3. FYDP Overview

FYDP Title: AI Based Body Measurement App

Sr. No	Roll Numbers	Name	Signatures
1.	BCS22014	Hassan Abdullah	
2.	BCS22044	H Muhammad Ertaza	
3.	BCS22045	Muhammad Imran	
4.	BCS22046	Dawood Rasheed	

Table 1 Project Proposal Summary

FYDP Goals
1. To achieve highly accurate body measurement extraction using Artificial Intelligence. 2. To reduce reliance on cloud-based processing and enhance user data privacy through edge computing. 3. To provide a customized and intelligent online dress design and shopping experience.

FYDP Objectives
1. Develop an AI-based measurement solution utilizing MoveNet / ML Kit for precise body measurement extraction from static images. 2. Implement model inference and measurement computation locally on the user's device to ensure privacy and real-time processing. 3. Enable users to browse and select dress designs tailored to their calculated body measurements, ensuring personalized recommendations.
FYDP Success Criteria
1. Measurement accuracy of **90%.** 2. **100% data privacy** ensured through local (on-device) model execution. 3. Successful implementation of personalized dress design recommendations based on accurate measurements.
Assumptions:
1. Users possess smartphones equipped with functional cameras and stable internet connections. **2**. Devices support the required OS versions and SDKs for MoveNet / ML Kit pose detection. **3**. Users maintain correct posture and adequate lighting conditions during measurement capture. **4**. Firebase services remain accessible and stable for authentication and backend data management.
Risks & Obstacles
1. Application performance depends on device camera quality and computational capabilities. 2. Measurement precision may decrease under poor lighting or incorrect posture conditions. 3. The AI model currently supports **single-person** body measurement only. 4. The system does **not include payment gateway integration or delivery tracking**, which may be considered for future development.
Organization Address: University of the Punjab, Gujranwala Campus, Pakistan
Target End Users: Online Shoppers, E-commerce Platforms, Tailors, and Fitness Enthusiasts
Suggested Project Supervisor: Dr. Sulman Nasir
Approved By:
Date: November 10, 2025

4. Tools, Libraries and Technologies with Reasoning

Table 2 Tools and Technologies

Category	Name	Version	Rationale
Tools	Android Studio	(latest stable version)	Integrated Development Environment (IDE) for building the application, leveraging the Flutter framework.
Tools	Visual Studio Code (VS Code)	(latest stable version)	Lightweight editor for Flutter and web development, useful for front-end web interface integration.
Tools	Chrome DevTools	(latest stable version)	Essential for debugging and performance testing of the Flutter Web interface.
Tools	Firebase	(Relevant versions for core services)	Comprehensive backend services for secure user authentication, real-time database, and cloud storage.
Libraries	TensorFlow Lite (TFLite)	^0.10.4 / ^2.0.1	Used for executing the machine learning model locally on the user's device (Edge Computing) to perform real-time body keypoint detection and inference.
Libraries	ML Kit (Google)	(Integrated via SDK)	A preferred model for enhanced on-device pose detection and measurement extraction.
Libraries	Flutter Packages (e.g., camera, image_picker)	(Various versions)	Essential third-party libraries for core functionalities such as real-time camera feed, image handling, and UI enhancements.
Technology	Flutter Framework	v3.x	Cross-platform development framework used to build the interactive and responsive app front-end with a unified codebase.
Technology	Edge Computing	N/A	Leverages the user's device's computational capabilities to perform ML inference locally, improving responsiveness, user privacy, and reducing latency.

5. Work Division

Table 3 Project Team Member Work Division

Sr. No	Roll Number	Name	Role Assignment & Work Division
1.	BCS 22045	Muhammad Imran	**Project Lead Backend & Database Engineer**: Focus on Firebase Integration (Authentication, Real-Time Database, Cloud Storage) and managing data security and logging.
2.	BCS 22044	Hafiz Muhammad Ertaza	**System Developer & AI Model Integrator**: Focus on implementing the core Pose Detection Module (TFLite/ML Kit), algorithms, and ensuring overall system stability.
3.	BCS 22014	Hassan Abdullah	**Frontend & UI/UX Developer**: Focus on designing and implementing the User Interface (UI), including image upload, measurement display, and dress visualization.
4.	BCS 22046	Dawood Rasheed	**QA Specialist & Testing Lead**: Focus on Requirements Analysis, designing and executing Test Cases (e.g., Accuracy Validation), and Performance Optimization across devices.

6. References

Books / Conference Proceedings

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